SArkhypov-Student Project Kernel-2025

by SERGII ARKHYPOV

Submission date: 02-Jan-2026 03:28PM (UTC+0000)
Submission ID: 271915139
File name: Arkhypov-Student-Project_Kernel-2025.pdf (135.68K)
Word count: 542
Character count: 3099

School of Computing and Mathematical Sciences

Project Kernel - Student Idea

INSTRUCTIONS FOR STUDENTS

(1) complete boxes A, B
(2) send the form electronically **to potential supervisor (see Instructions for Supervisor)**
If no supervisor is named on the submitted form, the project coordinator will assign you one
(3) Upload the completed form to Moodle by deadline specified

INSTRUCTIONS FOR SUPERVISORS

(1) complete boxes C and D, specifying whether the project is appropriate for the student's degree
(2) Return completed form to student who should upload it on Moodle

A

Proposal from: Sergii Arkhypov
Email: sarkhypov@gmail.com
MSc Programme of study: Data Analytics and AI
Student ID: 14019708 (B302213)

B

Suggested working title:

Machine Learning for Modeling High Dimensional Joint Dependency

General topic area (e.g. Data Analytics, Databases, Machine Learning, Software Development):

Data Analytics and Machine Learning

Brief description of problem to be tackled and up to three key references:

Despite the popularity of ML/AI algorithms in finance, they have mainly been used for either one or a small number of random factors. Nevertheless, joint analytics of high dimensionality (e.g., >1000 random factors) present serious challenges. This task is hard to address with classical statistical methods, and the available toolset is quite limited, i.e., mainly based on Gaussian copulas. There is hope that modern advances in ML/AI algorithms could help expand this toolset and provide alternative approaches for modeling high-dimensional joint dependency.

The main aim of this work would be to develop a machine learning algorithm for calibrating a high-dimensional copula with tail dependence different from Gaussian, while still reflecting the observed history with regards to selected measures (i.e., matching correlation matrix, matching pairwise tail dependence at the specific percentile, etc.). The latter is expected to require the creation of a custom loss function to achieve minimization with regards to those measures. On the simulation part, a range of different approaches can be considered, starting from semi-analytical methods to Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs).

References:

D.Oh, A.Patton (2015) - "Modelling dependance in High dimensions with factor copulas" (link)
R.Cont et al. (2022) - "Tail-GAN: Nonparametric scenario generation for tail risk estimation" (link)
H.Buhler, B.Horvath (2021) - "A data-driven market simulator for small data environments" (link)

Level of project difficulty (from 1 for "appropriate to the degree" to 5 "for very challenging")**: 3**

For this project the student is expected to be aware of the following material: (list concepts or relevant modules**)**

BBK modules:
Applied Machine Learning
Data Science Techniques and Applications
Neural Networks and Deep Learning

Other important concepts:

Probability theory
Random processes
Sklar's theorem and copula theory

Resources required: (list dataset, software and hardware)
Financial data available at yahoo finance (e.g.. Equity stock prices, FX spot rates)
VS Code IDE
Python language and corresponding packages (e.g. Keras, Tensorflow, Scikit learn etc.)

Are these currently available within the School? **YES**

Is there an outside company involved? **NO**

C This project is approved for Degrees: **AC CS DS IT AS**
(project supervisor to delete as necessary)

D (project supervisor to complete)
I am prepared to supervise the above student and project.

Name (Block capitals) **Date**

E (project coordinator) **Status:** **OK** (accepted) **RP** (revise proposal) **RA** (revision accepted)
Supervisor: *Dr Alessandro Provetti*
Comments:

Date

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GRADEMARK REPORT

FINAL GRADE

GENERAL COMMENTS

100/100

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Text Comment. Dr Alessandro Provetti